

ALICANTE AP, Spain

WMO: 083600

Lat: 38.283N

Lon: 0.571W

Elev: 43

StdP: 100.81

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.4	4.8	-7.7	2.0	8.8	-5.2	2.4	10.4	12.9	14.4	11.4	14.3	2.6	290	0.398

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.3	32.8	21.2	31.4	21.5	30.5	21.7	25.2	28.7	24.5	28.4	23.8	28.0	4.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.1	27.2	23.2	18.1	27.0	22.4	17.2	26.8	77.2	28.4	74.1	28.4	71.5	27.9	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.3	9.0	7.9	DB	0.6	36.7	1.3	1.7	-0.3	37.9	-1.0	38.9	-1.7	39.9	-2.7	41.2	
			WB	-1.6	25.6	1.6	1.1	-2.7	26.4	-3.7	27.0	-4.6	27.7	-5.8	28.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.5	12.0	12.3	14.1	16.2	19.3	23.2	25.8	26.4	23.7	20.2	15.4	12.6
	DBStd	5.66	2.67	2.76	2.72	2.22	2.23	2.14	1.63	1.55	2.08	2.34	2.91	2.53
	HDD10.0	28	10	8	2	0	0	0	0	0	0	0	1	7
	HDD18.3	872	198	171	136	71	16	0	0	0	9	96	177	
	CDD10.0	3114	71	72	128	186	287	394	491	508	411	315	163	89
	CDD18.3	917	1	1	4	6	44	145	233	249	161	66	7	0
	CDH23.3	7589	1	3	26	37	223	1064	2254	2514	1146	296	24	1
	CDH26.7	2242	0	1	4	3	33	272	746	862	277	42	2	0
Wind	WSAvg	3.2	3.6	3.6	3.5	3.4	3.1	3.0	3.0	2.9	3.0	2.8	3.4	3.4
Precipitation	PrecAvg	280	22	19	25	31	25	12	4	9	39	34	32	28
	PrecMax	467	74	92	71	81	100	74	22	63	190	150	88	151
	PrecMin	140	0	0	0	4	0	0	0	0	2	2	4	1
	PrecStd	79	21	22	20	22	23	16	5	15	40	30	24	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.2	23.4	26.2	26.5	29.6	33.4	34.1	35.0	33.2	30.0	26.0	22.1
		MCWB	13.1	13.4	14.6	15.6	17.4	18.9	20.6	21.4	20.2	19.6	16.7	14.4
	2%	DB	20.1	21.0	23.2	24.1	27.2	31.0	32.3	32.8	30.8	27.7	23.5	20.1
		MCWB	12.3	12.8	13.7	14.7	16.9	19.6	21.3	22.1	20.6	19.0	15.8	13.3
	5%	DB	18.4	19.2	21.2	22.7	25.7	29.5	31.1	31.3	29.4	26.2	21.9	18.7
		MCWB	11.4	11.8	13.0	14.4	16.7	19.5	21.5	22.5	21.1	18.9	15.4	12.7
	10%	DB	16.9	17.5	19.5	21.2	24.2	28.1	30.1	30.3	28.1	24.9	20.3	17.3
		MCWB	11.1	11.3	12.4	13.9	16.6	19.3	21.8	22.7	21.0	18.5	14.6	12.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.8	15.0	16.2	17.7	20.5	23.4	25.7	26.3	25.0	22.0	18.8	16.1
		MCDB	18.8	20.7	22.4	22.8	23.9	28.0	29.0	29.2	27.9	26.1	22.6	19.3
	2%	WB	13.6	13.8	15.1	16.6	19.4	22.4	24.7	25.6	23.7	21.0	17.6	14.9
		MCDB	17.7	18.6	20.5	21.1	23.4	26.6	28.5	28.8	27.6	24.8	21.4	18.0
	5%	WB	12.7	13.0	14.2	15.8	18.6	21.6	24.1	24.9	22.9	20.4	16.6	13.9
		MCDB	16.8	17.3	18.9	20.3	22.8	25.9	28.1	28.6	26.9	24.3	20.3	17.3
	10%	WB	11.9	12.1	13.6	15.1	17.9	21.0	23.4	24.2	22.2	19.7	15.7	13.0
		MCDB	16.0	16.3	17.9	19.6	22.3	25.5	27.8	28.3	26.4	23.6	19.3	16.5
Mean Daily Temperature Range	5% DB	MDBR	9.5	9.7	10.1	9.7	9.3	9.1	8.5	8.3	8.7	9.0	9.0	9.1
		MCDBR	11.1	11.7	12.6	11.7	11.5	11.2	10.2	10.0	10.4	10.8	10.4	10.4
		MCWBR	6.2	6.2	6.2	5.7	5.3	5.1	5.3	5.3	5.4	5.5	5.8	6.0
	5% WB	MCDBR	9.7	10.5	10.7	10.2	9.4	9.0	8.3	8.3	8.8	9.1	9.1	9.3
		MCWBR	6.2	6.5	6.3	5.7	5.7	4.9	5.0	5.0	5.1	5.1	5.3	5.6
Clear-Sky Solar Irradiance	taub	0.311	0.336	0.382	0.394	0.421	0.442	0.449	0.452	0.423	0.388	0.344	0.319	
	taud	2.502	2.453	2.329	2.318	2.270	2.239	2.233	2.230	2.313	2.399	2.482	2.503	
	Ebn at Noon	872	891	876	884	862	841	832	820	823	818	823	835	
	Edh at Noon	79	94	117	126	134	139	138	136	118	98	80	74	
All-Sky Solar Radiation	RadAvg	2.50	3.30	4.50	5.54	6.60	7.44	7.36	6.37	4.96	3.66	2.59	2.15	
	RadStd	0.25	0.34	0.37	0.40	0.45	0.23	0.19	0.21	0.27	0.29	0.24	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	-1.65	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)	
(47)	Regional (5 neighbors)	N/A	N/A	-1.35	+0.41	N/A	N/A	N/A	N/A	N/A	+49	(47)	

Nomenclature: See separate page